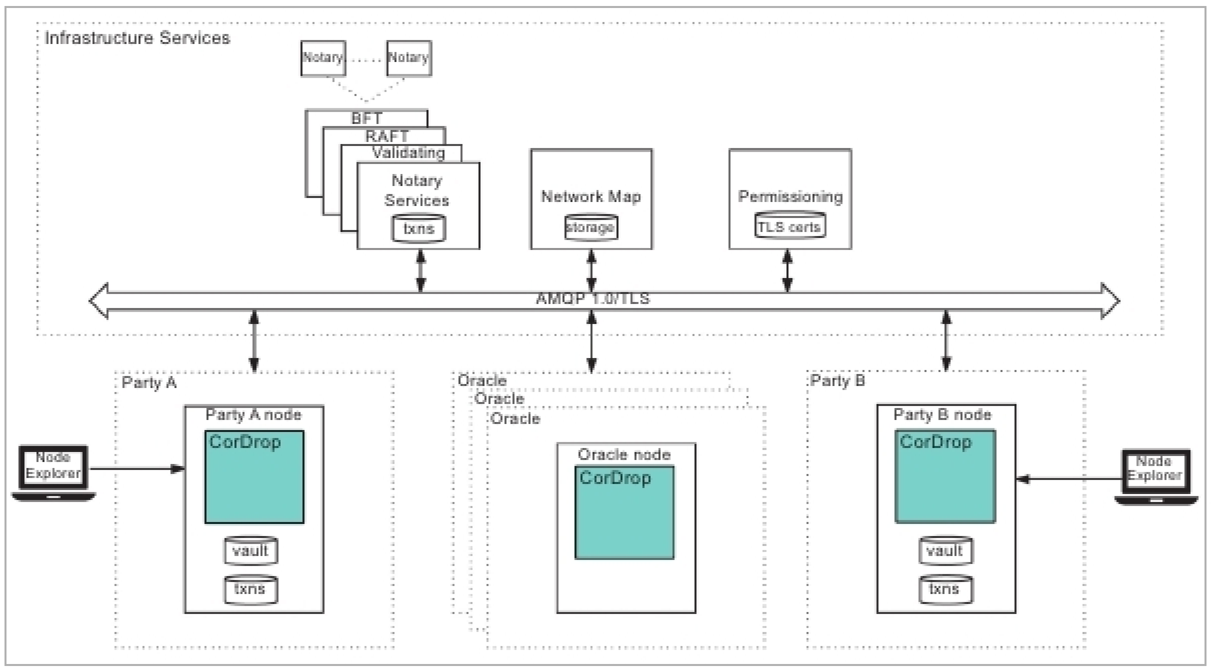

Figure 3: The components of Corda

Hyperledger

Hyperledger is an open source collaborative effort hosted by the Linux Foundation and doesn't support Bitcoin or any other cryptocurrency. The objective of Hyperledger is to advance cross-industry collaboration by developing blockchains and distributed ledgers, with a focus on improving the performance and reliability of these systems as compared to traditional cryptographic designs – in order to make them capable of supporting global business transactions by major technological, finance and supply chain companies.

Figure 2 highlights the Hyperledger reference architecture.

The following are some of the Hyperledger frameworks.

- ***Hyperledger Sawtooth:*** This is a modular blockchain suite developed by Intel, which makes use of the consensus algorithm called 'Proof of Elapsed Time (PoeT)'.
- ***Hyperledger Iroha:*** This is a project of some Japanese companies to create an easy-to-implement framework for a blockchain.
- ***Hyperledger Fabric:*** This has been designed by IBM as a plug-and-play implementation of blockchain technology for designing high-scale blockchain applications with a high degree of flexible permissions.
- ***Hyperledger Burrow:*** This develops a permissible smart contract machine along the specifications of Ethereum.

Some of the popular tools of Hyperledger are listed below.

- ***Caliper*****:** This is a benchmarking tool that allows users to measure the performance of specific blockchain implementations with a set of predefined use cases.
- ***Cello*****:** This blockchain module toolkit brings the on-demand 'as-a-service' deployment model to the blockchain ecosystem to reduce the effort of managing and destroying blockchains.
- ***Composer*****:** This creates smart contracts and applications to solve complex business problems.
- ***Explorer*****:** This is a user interface to view, deploy and query blocks, transactions, chain codes as well as other relevant information in the ledger.
- ***Quilt:*** This is a payment protocol developed to transfer value across distributed ledgers and non-distributed ledgers.

Official website**:** *https://www.hyperledger.org/*

Corda

Corda is an open source distributed ledger platform for recording and processing financial agreements that support smart contracts. The objective of Corda is to provide a platform with common services to ensure that any services built on top are compatible with network participants and innovations happen faster.

Corda is the only platform offering the 'universal interoperability of public networks with the privacy of private networks'. As compared to traditional blockchains, Corda ensures minimal information leaks by sharing transaction data only with participants that require it. It maintains confidentiality and provides encryption to client data.

As a smart contract platform and host of distributed apps called Dapps, it acts as a gateway to a network of fully interoperable Dapps for finance and commerce. These are called CorDapps.

Listed below are some of the key components of Corda.

- ***Node*****:** Every node hosts Corda services and executes CorDapps using JVM and communicates using AMQP/1.0 over TLS
  - Permissioning service
  - Network map service
  - Pluggable notary service types
  - Oracle services
  - CorDapps
  - Corda applications

Official website: *https://www.corda.net/*
Latest version: 3.1

HydraChain

HydraChain is an open source blockchain platform designed by Brainbot Technologies and the Ethereum Project. It is regarded as an extension to the Ethereum blockchain platform, and provides support to create private blockchain networks.

It is fully compatible with API level as well as contract level protocols in Ethereum, and smart contracts are designed using the Python programming language. In a HydraChain network, all the blocks are not allowed to enter the network without proper validation, and any block can only be added with a proper validator's signature. Once the block enters the network, it becomes persistent with no reverts.